University of Sunderland

Faculty: Programme

NAME:
MODULE CODE:
MODULE TITLE:
MODULE LEADER:
SUBMISSION DATE AND TIME:
ASSIGNMENT:

Academic Misconduct is an offence under university regulations, and this involves:

- **Plagiarism** – where you use information from another information source (including your previously submitted work) and pass it off as your own. This can be through direct copying, poor paraphrasing and/or absence of citations.
- **Collusion** – where you work too closely, intentionally, or unintentionally, with others to produce work that is similar in nature. This can be through loaning of materials, drafts or through unauthorised use of a fellow student's work.
- **Asking another person to write your assignment** – where you ask another individual or company to complete your work for you, be that paid or unpaid, and submit it as if it were your own.
- **Unauthorised use of artificial intelligence** – where you use artificial intelligence tools to generate your assignment instead of completing it yourself and/or where you have not been given permission to use artificial intelligence tools by your module leader. Please complete the following declaration around you use of artificial intelligence tools in your assignment.

STATEMENT ON USE OF ARTIFICIAL INTELLIGENCE TOOLS:

• I have used artificial intelligence tools to generate an idea for my assignment:	**YES/NO**
• I have used artificial intelligence tools to write my assignment for me:	**YES/NO**
• I have used artificial intelligence tools to brainstorm ideas for my assignment:	**YES/NO**
• I have used artificial intelligence tools to correct my original assignment:	**YES/NO**

DECLARATION

- I understand that by submitting this piece of work I am declaring it to be my own work and in compliance with the university regulations on Academic Integrity.
- I confirm that I have done this work myself without external support or inappropriate use of resources.
- I understand that I am only permitted to use artificial intelligence tools in line with guidance provided by my Module Leader, and I have not used artificial intelligence tools outside this remit.
- I confirm that this piece of work has not been submitted for any other assignment at this or another institution prior to this point in time.
- I can confirm that all sources of information, including quotations, have been acknowledged by citing the source in the text, along with producing a full list of the sources used at the end of the assignment.
- I understand that academic misconduct is an offence and can result in formal disciplinary proceedings.
- I understand that by submitting this assignment, I declare myself fit to be able to complete the assignment and I accept the outcome of the assessment as valid and appropriate.

Please read **all** instructions and information carefully.
This assignment contributes 70% to your final module mark and will assess the following learning outcomes:

- Ability to apply software engineering tools and techniques in the testing and debugging of a computer program (Learning Outcome 2).
- Ability to apply appropriate processes and articulate the rationale for design decisions throughout the process of developing and documenting a computer program (Learning Outcome 3).

Module code:	CET131 Software Engineering
Module assessor:	Dr Sardar Jaf
Title of assessment:	Software Engineering Assignment 2 Software Implementation Task

Important Information

You are required to submit your work within the bounds of the University Infringement of Assessment Regulations (see your Programme Guide). Plagiarism, paraphrasing and downloading large amounts of information from external sources, will not be tolerated and will be dealt with severely. Although you should make full use of any source material, which would normally be an occasional sentence and/or paragraph (referenced) followed by your own critical analysis/evaluation. You will receive no marks for work that is not your own. Your work may be subject to checks for originality which can include use of an electronic plagiarism detection service.

For this assessment you are asked to submit an individual piece of work, therefore the work must be entirely your own. The safety of your assessments is your responsibility. You must not permit another student access to your work.

Referencing for this assessment should be done using the ***Harvard referencing system****. (see your Programme Guide).*

Please ensure that you retain a duplicate of your assignment. We are required to send samples of student work to the external examiners for moderation purposes. It will also safeguard in the unlikely event of your work going astray.

Submission instruction:

Submission Date and Time	Check Canvas submission page for date and time of submission
Submission Location	Digital copy via Canvas
Document Format	1. A fully documented (i.e. containing suitable source comments) software program which should implement the system requirements using any programming language you are competent in using it. 2. Screencast. Your screencast should be in suitable file format such as MPEG or MOV.
Assignment weight	70%
Other requirements	You must adhere to the above assessment requirements. Your source code will be checked for plagiarism.

CET131 Assessment 2

Scenario

You have recently been employed as a graduate software developer working remotely for a new technology consultancy company. After completing a number of induction-based tasks, your line manager has issued you with your first client-based project.

WeAreCars are a new car rental company based in Sunderland. They are needing a software solution to assist in the rental of their vehicles. This solution needs to be a standalone windows forms based program to ensure future compatibility with legacy systems. The company has strict initial requirements and are on a tight budget due to them being a new company. However, WeAreCars wish to ensure the software is to be of the highest standard and do not want to launch the solution until all requirements are fully implemented and tested.

System Specification requirements (80%)

WeAreCars would like a software application that will assist in the rental of cars. You are you required to implement a software prototype to include the features below using any programming language you prefer. The system should have user interface as console or graphical format to allow users to interact with the system.

System features:

- A welcome splash screen. This is to display a welcome message to the user and brief instructions on how to use the interface.
- A staff login page that allows a staff member to login into the system, using the following fixed username and password
    - Username : sta001
    - Password: givemethekeys123
- An area that displays a list of the rented cars
- A main booking area that will process the booking. This area will allow a staff member to enter and/or select certain options (Mandatory fields are denoted with a *):
    - Customer First Name*
    - Customer Surname*
    - Customer Address*
    - Customer age*
    - Valid driving license*
        - If no, the booking can't proceed
    - Number of days* (£25 per day)
        - Between 1 and 28
    - Type of car selected*
        - City car (No extra charge)
        - Family car (+ £50)

        - Sports car (+ £75)
        - SUV (+ £65)
    - Fuel type*
        - Petrol (No extra charge)
        - Diesel (No extra charge)
        - Hybrid (+ £30)
        - Full electric (+ £50)
    - Optional extras
        - Unlimited mileage (+ £10 per day)
        - Breakdown cover (+ £2 per day)
    - A summary screen that will confirm the data entered, options selected and a total price, before storing the booking.
- The system should validate input data and display appropriate error messages.

WeAreCars would like this system to be as robust and as user friendly as possible. Attention to detail and the inclusion of validation and help systems / tooltips are recommended.

System Demonstration (20%)

Prepare up to 10 minutes of recorded presentation of all the features and functionalities that you have implemented for this project. Your presentation should:

- Have a clear view of the source code.
- Show all the features you have implemented according to the system specifications outline in the task above.
- You should verbally describe the system features during the demonstration.

Marking Scheme

Criteria	Fail (0-14)	Fail (15-29)	Fail (0-39)	Fail (30-39)	Pass (40-49)	pass (50-59)	pass (60-69)	pass (70-79	pass (79-80)	pass (80-89)	pass (90-100)
System features/functionality (60%)	Irrelevant functionalities are implemented	Barely relevant functionalities are implemented with some deficiencies.	Some relevant functionalities implemented but with major deficiencies.	Some relevant functionalities implemented but with minor deficiencies.	Some relevant functionalities are implemented with no deficiencies.	Most of the functionalities are implemented but with major deficiencies	Most of the functionalities are implemented but with minor deficiencies	Most of the functionalities are implemented with no deficiencies.	All the functionalities implemented, with major deficiencies and including extra features	All the functionalities implemented with minor deficiencies and including extra features	All the functionalities implemented and with no deficiencies and including extra features
Source code annotation and naming convention (20%)	No source code annotation	Very little source code annotation	Some source code annotation with major deficiencies	Some source code annotation with minor deficiencies	Some source code annotation with no deficiencies	Good source code annotation with major deficiencies	Good source code annotation with minor deficiencies	Good source annotation code with no deficiencies	Full source code annotation but major deficiencies	Full source code annotation but manor deficiencies	Full source code annotation but no deficiencies
Demonstration (20%)	No coherent structure and organisation. Some irrelevant or poorly presented content	Relevant, little coherent structure and organisation. Some relevant and acceptable presentation of content	Relevant, some coherent structure and organisation. Relevant content but there are major inconsistencies. Or sometime poorly presented content	Relevant, some coherent structure and organisations. Relevant content but with some minor inconsistencies. Or sometime poorly presented content	Relevant, largely coherent and logically structured but with some minor inconsistency or poorly presented content	Relevant, largely coherent and logically structured with no inconsistency or poorly presented content	Relevant, coherent and logically structured with minor inconsistency or poorly presented content	Relevant, coherent and logically structured with no inconsistency or poorly presented content	Excellent and relevant, content, coherent and logically structured. Excellent use of headings and titles. no inconsistency and clearly presented content	Relevant, coherent and logically structured with no inconsistency or poorly presented content	Excellent and relevant, content, coherent and logically structured. Excellent use of headings and titles. no inconsistency and clearly presented content